Annex IV. Debt Sustainability Analysis

Figure 1. Chile: Macro-Fiscal Stress Tests

(in percent of GDP unless otherwise indicated)

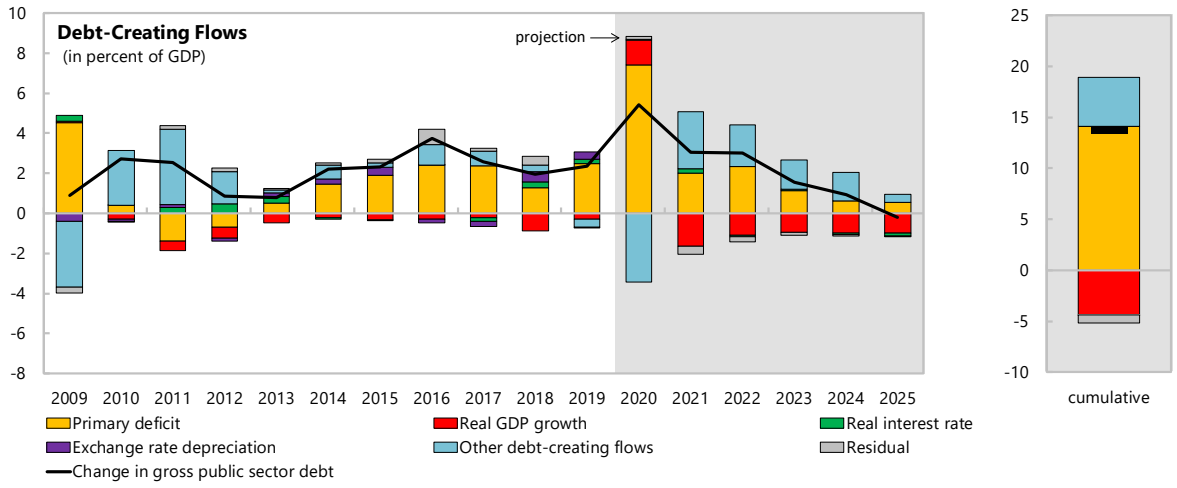
Debt, Economic and Market Indicators [1/]

	Actual			Projections					
	2009-2017 [2/]	2018	2019	2020	2021	2022	2023	2024	2025
Nominal gross public debt	14.1	25.6	27.9	33.3	36.4	39.4	41.0	41.9	41.7
Public gross financing needs	3.0	4.9	4.9	8.5	4.8	3.0	3.5	1.2	1.8
Real GDP growth (in percent)	3.0	3.9	1.1	-4.5	5.3	3.2	2.5	2.5	2.5
Inflation (GDP deflator, in percent)	4.4	2.4	2.7	3.9	1.9	2.7	2.8	2.8	2.8
Nominal GDP growth (in percent)	7.5	6.4	3.8	-0.8	7.3	6.0	5.4	5.4	5.4
Effective interest rate (in percent) [4/]	6.3	3.8	3.7	3.8	2.6	2.6	3.0	2.6	2.5

As of May 14, 2020		
Sovereign Spreads		
EMBIG (bp) 3/		271
5Y CDS (bp)		114
Ratings	Foreign	Local
Moody's	A1	A1
S&Ps	A+	AA-
Fitch	A	A+

Contribution to Changes in Public Debt

	Actual			Projections							
	2009-2017	2018	2019	2020	2021	2022	2023	2024	2025	cumulative	debt-stabilizing
Change in gross public sector debt	2.1	2.0	2.3	5.4	3.1	3.0	1.6	0.9	-0.2	13.7	primary
Identified debt-creating flows	1.9	1.5	2.4	5.3	3.4	3.2	1.7	1.0	-0.2	14.5	balance [9/]
Primary deficit	1.3	1.3	2.5	7.4	2.0	2.3	1.1	0.6	0.5	14.1	-0.7
Primary (noninterest) revenue and grants	20.6	21.5	20.7	18.5	21.5	21.0	21.3	21.2	21.1	124.5	
Primary (noninterest) expenditure	21.8	22.8	23.2	25.9	23.5	23.3	22.5	21.8	21.7	138.6	
Automatic debt dynamics [5/]	-0.2	0.0	0.3	1.3	-1.4	-1.2	-0.9	-1.1	-1.1	-4.4	
Interest rate/growth differential [6/]	-0.2	-0.6	0.0	1.3	-1.4	-1.2	-0.9	-1.1	-1.1	-4.4	
Of which: real interest rate	0.1	0.3	0.2	0.0	0.2	-0.1	0.0	-0.1	-0.1	0.0	
Of which: real GDP growth	-0.3	-0.9	-0.3	1.3	-1.6	-1.1	-0.9	-1.0	-1.0	-4.4	
Exchange rate depreciation [7/]	0.0	0.5	0.4	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.9	0.3	-0.4	-3.4	2.9	2.1	1.5	1.4	0.4	4.8	
Net acquistion of Financial Assets (negative)	0.3	0.0	-0.6	-3.6	2.7	1.9	1.4	1.4	0.4	4.2	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Net Repayment of Recognition Bond	0.6	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.0	0.7	
Residual, including asset changes [8/]	0.1	0.4	0.0	0.1	-0.4	-0.2	-0.2	-0.1	0.0	-0.8	



Source: IMF staff.

1/ Public sector is defined as central government.

2/ Based on available data.

3/ EMBIG.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

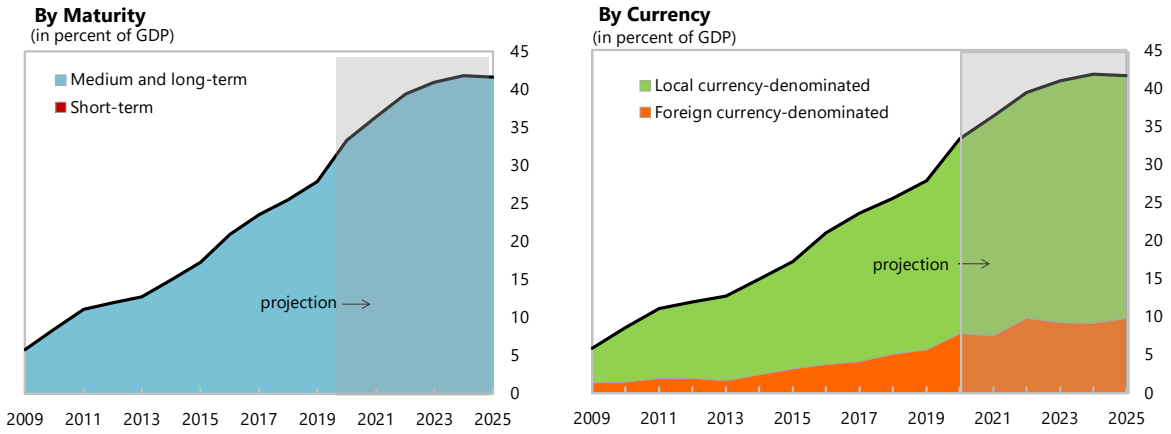
8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

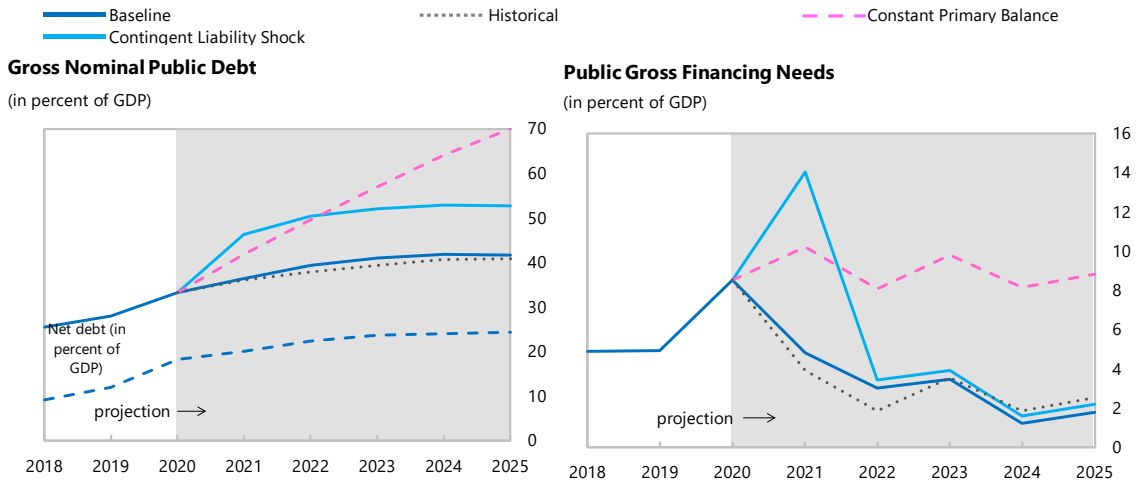
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Figure 2. Chile: Composition of Public Debt and Alternative Scenarios

Composition of Public Debt



Alternative Scenarios



Underlying Assumptions

(in percent)

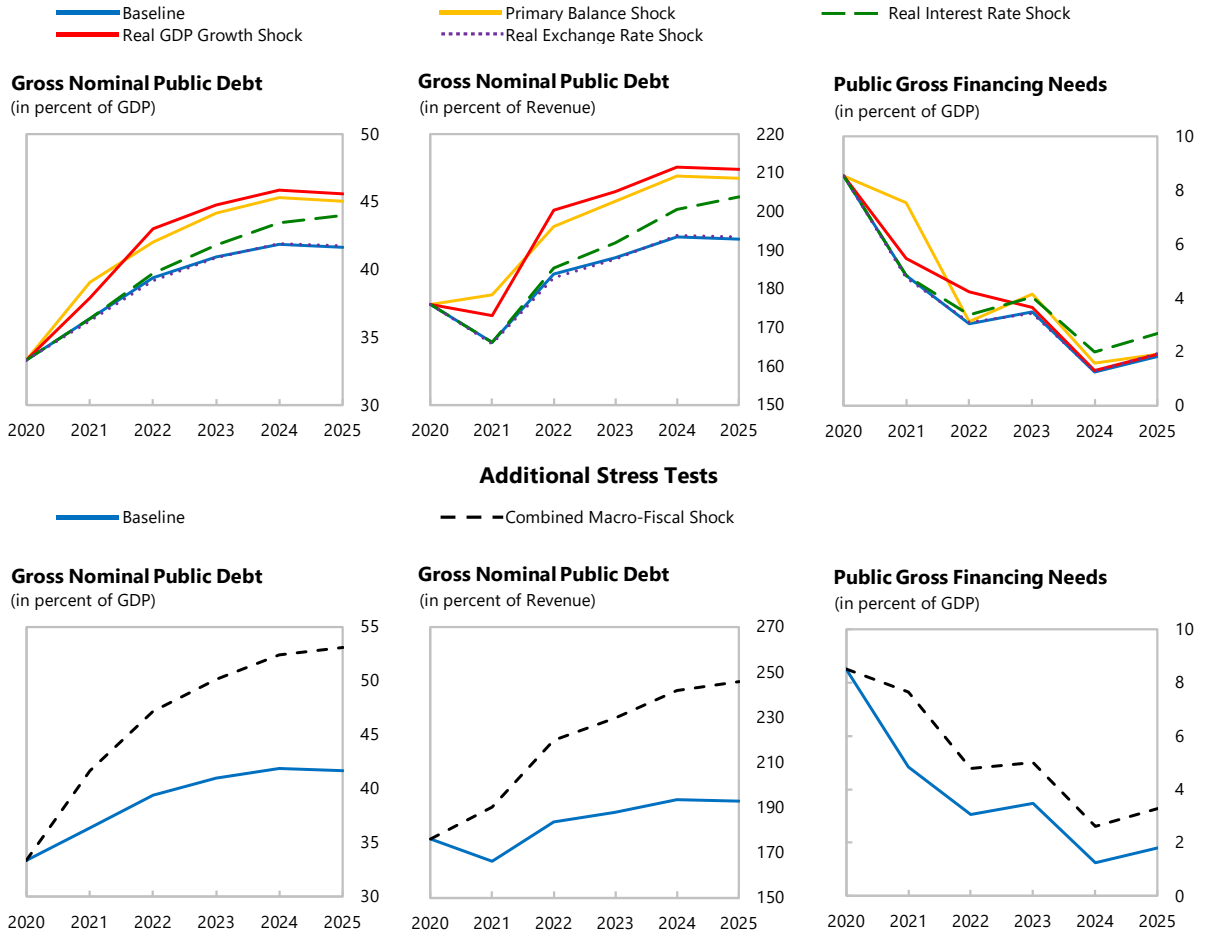
Baseline Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-4.5	5.3	3.2	2.5	2.5	2.5
Inflation	3.9	1.9	2.7	2.8	2.8	2.8
Primary Balance	-7.4	-2.0	-2.3	-1.1	-0.6	-0.5
Effective interest rate	3.8	2.6	2.6	3.0	2.6	2.5
Constant Primary Balance Scenario						
Real GDP growth	-4.5	5.3	3.2	2.5	2.5	2.5
Inflation	3.9	1.9	2.7	2.8	2.8	2.8
Primary Balance	-7.4	-7.4	-7.4	-7.4	-7.4	-7.4
Effective interest rate	3.8	2.6	2.2	2.5	2.1	1.9

Historical Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-4.5	3.3	3.3	3.3	3.3	3.3
Inflation	3.9	1.9	2.7	2.8	2.8	2.8
Primary Balance	-7.4	-1.1	-1.1	-1.1	-1.1	-1.1
Effective interest rate	3.8	2.6	2.9	3.5	3.2	3.1
Contingent Liability Shock						
Real GDP growth	-4.5	3.3	1.3	2.5	2.5	2.5
Inflation	3.9	1.4	2.2	2.8	2.8	2.8
Primary Balance	-7.4	-11.0	-2.3	-1.1	-0.6	-0.5
Effective interest rate	3.8	3.0	2.9	3.2	2.8	2.7

Source: IMF staff.

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Figure 3. Chile: Macro-Fiscal Stress Tests



Underlying Assumptions

(in percent)

Primary Balance Shock	2020	2021	2022	2023	2024	2025
Real GDP growth	-4.5	5.3	3.2	2.5	2.5	2.5
Inflation	3.9	1.9	2.7	2.8	2.8	2.8
Primary balance	-7.4	-4.7	-2.3	-1.7	-0.9	-0.6
Effective interest rate	3.8	2.6	2.6	2.9	2.6	2.5
Real Interest Rate Shock						
Real GDP growth	-4.5	5.3	3.2	2.5	2.5	2.5
Inflation	3.9	1.9	2.7	2.8	2.8	2.8
Primary balance	-7.4	-2.0	-2.3	-1.1	-0.6	-0.5
Effective interest rate	3.8	2.6	3.5	4.4	4.5	4.5
Combined Shock						
Real GDP growth	-4.5	3.3	1.3	2.5	2.5	2.5
Inflation	3.9	1.4	2.2	2.8	2.8	2.8
Primary balance	-7.4	-4.7	-3.5	-1.7	-0.9	-0.6
Effective interest rate	3.8	2.8	3.6	4.4	4.5	4.6

Real GDP Growth Shock	2020	2021	2022	2023	2024	2025
Real GDP growth	-4.5	3.3	1.3	2.5	2.5	2.5
Inflation	3.9	1.4	2.2	2.8	2.8	2.8
Primary balance	-7.4	-2.6	-3.5	-1.1	-0.6	-0.5
Effective interest rate	3.8	2.6	2.6	3.0	2.6	2.5
Real Exchange Rate Shock						
Real GDP growth	-4.5	5.3	3.2	2.5	2.5	2.5
Inflation	3.9	7.3	2.7	2.8	2.8	2.8
Primary balance	-7.4	-2.0	-2.3	-1.1	-0.6	-0.5
Effective interest rate	3.8	2.8	2.6	3.0	2.7	2.5

Source: IMF staff.

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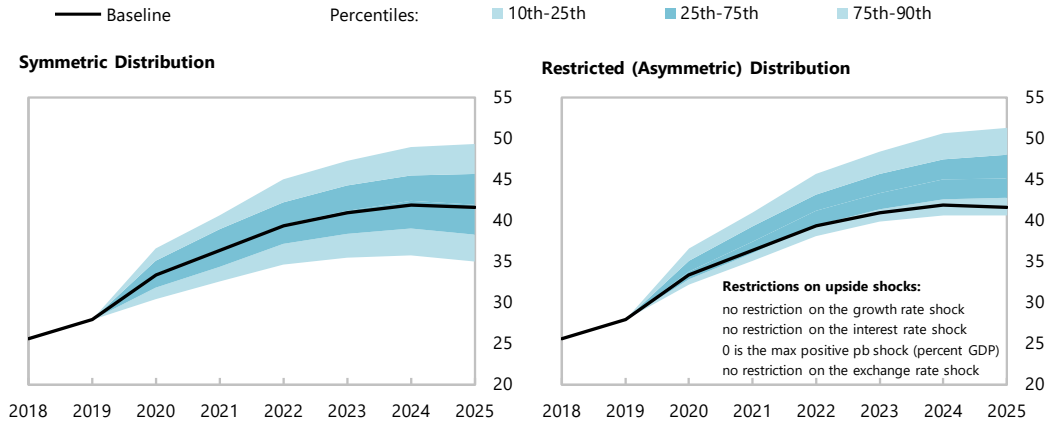
Figure 4. Chile: Risk Assessment

Heat Map

Debt level [1/]	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability shock
Gross financing needs [2/]	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability Shock
Debt profile [3/]	Market Perception	External Financing Requirements	Change in the Share of Short-Term Debt	Public Debt Held by Non-Residents	Foreign Currency Debt

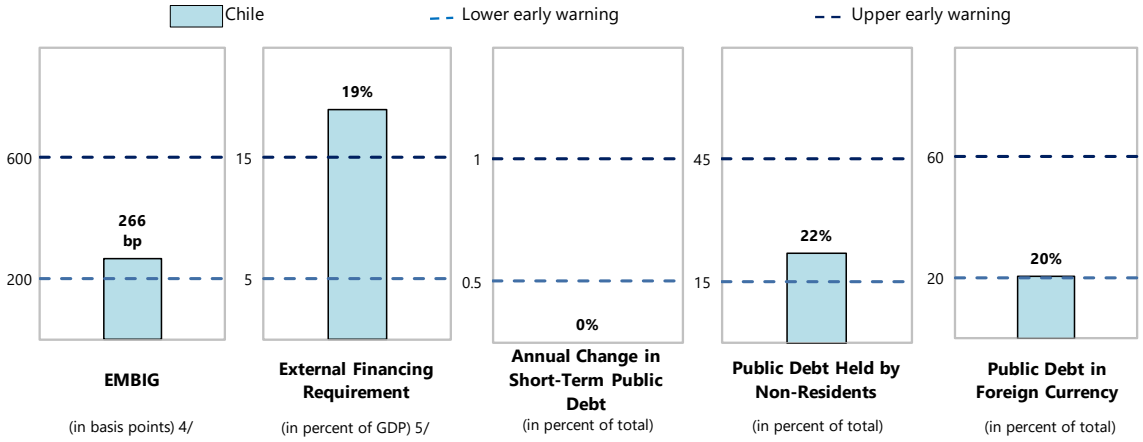
Evolution of Predictive Densities of Gross Nominal Public Debt

(in percent of GDP)



Debt Profile Vulnerabilities

(Indicators vis-à-vis risk assessment benchmarks, in 2019)



Source: IMF staff.

1/ The cell is highlighted in green if debt burden benchmark of 70% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.

2/ The cell is highlighted in green if gross financing needs benchmark of 15% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.

3/ The cell is highlighted in green if country value is less than the lower risk-assessment benchmark, red if country value exceeds the upper risk-assessment benchmark, yellow if country value is between the lower and upper risk-assessment benchmarks. If data are unavailable or indicator is not relevant, cell is white.

Lower and upper risk-assessment benchmarks are:

200 and 600 basis points for bond spreads; 5 and 15 percent of GDP for external financing requirement; 0.5 and 1 percent for change in the share of short-term debt; 15 and 45 percent for the public debt held by non-residents; and 20 and 60 percent for the share of foreign-currency denominated debt.

4/ EMBIG, an average over the last 3 months, 14-Feb-20 through 14-May-20.

5/ External financing requirement is defined as the sum of current account deficit, amortization of medium and long-term total external debt, and short-term total external debt at the end of previous period.

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